

Docker Networking & Volume Homework Tasks

Task 1: Docker Container Networking

- Create 3 containers:
 - Frontend
 - Backend
 - Database
- Use Nginx or Alpine images for the frontend and backend.
- Use the MySQL image for the database.
- Create 3 different Docker networks.
- Add the backend container to 2 networks.
- Check connectivity between the containers.

Task 2: Host Network

- Pull the Apache2 image from Docker Hub.
- Create an Apache2 container using the host network.
- Access the Apache website directly on port 80.

Task 3: Bind Mount

- Create a folder on your local machine.
- Create an `index.html` file with **Hello students** as the content.
- Bind mount the folder to an Nginx container.
- Access the Nginx website and verify the content.
- Modify the `index.html` file.
- Verify that the changes are reflected without restarting the container.

Task 4: Overlay Network

- Research Docker overlay networks.
- Understand their use cases.

- Understand how overlay networks work across multiple Docker hosts.

Submission

- Take screenshots of the completed exercises.
- Add the screenshots to the [README.md](#) file in your GitHub repository.
- Complete any remaining exercises from the session.